

Name _____



Preliminary HSC Final - 2013

Mathematics Extension 1

Time Allowed - 1 hour 30 minutes + 5minutes reading

Instructions: Calculators may be used in any parts of the task. For 1 Mark Questions, the correct answer is sufficient to receive full marks. For Questions worth more than 1 Mark, necessary working MUST be shown to receive full marks.

Multiple Choice	/6
Question 7	/12
Question 8	/12
Question 9	/12
Question 10	/12
Total	/54

Fill in the correct answer on the answer sheet - Questions 1 - 6 are worth 1 mark each

1. $P(x)$ is an even polynomial. When $P(x)$ is divided by $(x - 2)$ the remainder is 7. What is the remainder when $P(x)$ is divided by $(x + 2)$

A	-7	B	$-7x$
C	$7x$	D	7

2. By factorising $3^{n+1} + 3^n$, rewrite $\frac{3^{1001} + 3^{1000}}{4}$ as a power of 3

A	3^{1002}	B	3^{1000}
C	3^{1001}	D	3^{999}

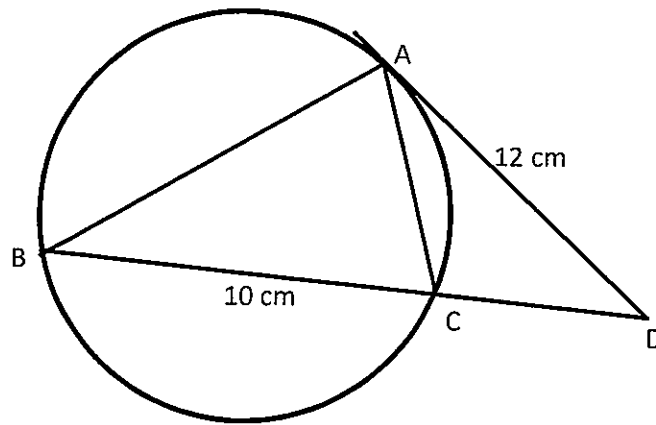
3. If $(x - 1)$ is a factor of $P(x) = 2x^3 + x^2 + 2x + a$, what is the value of a ?

A	-6	B	-5
C	5	D	6

4. Which of the following is an expression for $\sin(A + B) - \sin(A - B)$

A	$-2 \sin A \cos B$	B	$-2 \cos A \sin B$
C	$2 \cos A \sin B$	D	$2 \sin A \cos B$

5.



ABC is a triangle inscribed in a circle. The tangent to the circle at A meets BC produced at D, where $BC = 10\text{ cm}$ and $AD = 12\text{ cm}$. What is the length of CD?

A 6 cm

B 7 cm

C 8 cm

D 9 cm

6. How many arrangements of the word GEOMETRY are possible if the word TRY appears in the new word?

A $2 \times 6!$

B $2 \times 8!$

C $\frac{8!}{3!}$

D $\frac{1}{2} \times 6!$

- a) The point P divides the interval AB joining A $(-3, -4)$ and B $(1, 2)$ externally in the ratio 3:2. Find the coordinates of P. 3
- b) Find the acute angle between the lines $y = x + 2$ and $y = 2x - 3$.
(Give your answer to the nearest minute.) 3
- c) Consider the word QUADRILATERAL,
leaving answers in unsimplified factorial form find:
- (i) How many different arrangements of this word are possible? 1
 - (ii) In how many of these arrangements will both the L's be together and the R's be together? 1
 - (iii) In how many of the arrangements from part (i) will all the vowels be together? 1
- d) A committee of 4 boys and 3 girls is to be formed from a class of 15 boys and 10 girls.
- (i) How many different committees can be formed? 1
 - (ii) The class decides that Sue and Peter must not both be on the committee at the same time. How many committees are now possible? 2

- a) (i) Using long division, prove that $(2x - 3)$ is a factor of $P(x) = 6x^3 - 41x^2 + 24x + 36$.

2

- (ii) Hence completely factorise $P(x)$.

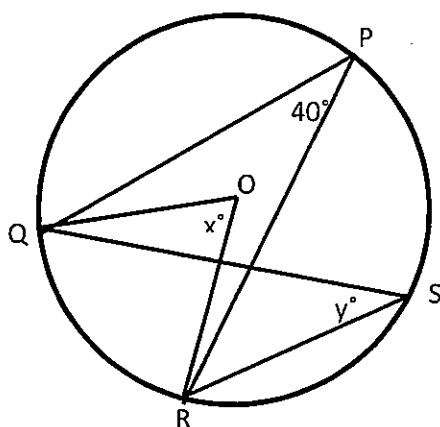
1

- b) Sketch the curve $y = (x + 1)^3(3 - 2x)^2$ showing all intercepts.

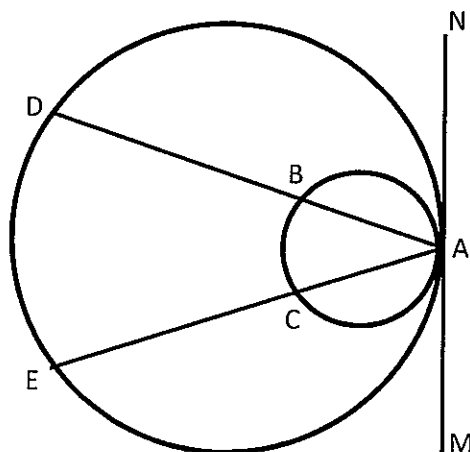
3

- c) P, Q, R, and S are on the circle with centre O. Angle $QPR = 40^\circ$. Find the values of x and y giving a reason for each answer.

2



d)



Two circles touch internally at A. MAN is the common tangent to the circles at A. ABD and ACE are two straight lines.

- (i) Show that $\triangle ABC$ is similar to $\triangle ADE$.

2

- (ii) Hence show that $\frac{AB \times AC}{AD \times AE} = \frac{BC^2}{DE^2}$.

2

Question 9 12 Marks (Begin a new sheet of paper)

Marks

a) (i) Write down the expansion of $\cos(A + B)$. 1

(ii) Hence find the exact value of $\cos 75^\circ$ 2

b) Use the substitution $t = \tan \frac{x}{2}$ to show that $\frac{1 + \cos x + \sin x}{1 - \cos x + \sin x} = \cot \frac{x}{2}$. 3

c) Use the substitution $t = \tan \frac{\theta}{2}$ to solve $\sqrt{3} \sin \theta - \cos \theta = 1$ for $0 \leq \theta \leq 360^\circ$ 3

d) (i) Express $\cos x + \sqrt{3} \sin x$ in the form $A \cos(x - \alpha)$ where $A > 0$ and α is acute. 2

(ii) Hence solve $\cos x + \sqrt{3} \sin x = 1$ for $0 \leq x \leq 360^\circ$ 1

Question 10 12 Marks **(Begin a new sheet of paper)**

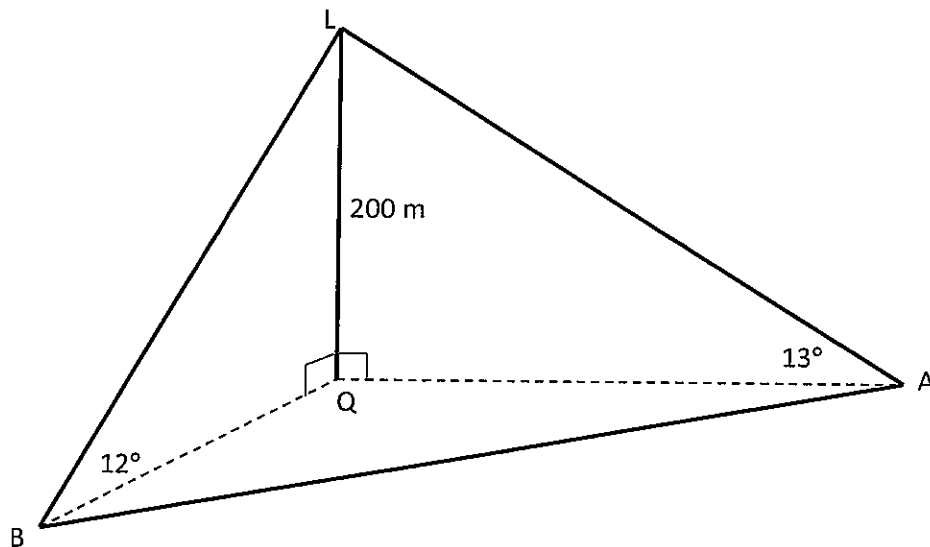
a) Solve the inequality $\frac{2x-1}{x} \geq x$ and graph your solution on a number line. 3

b) Let α, β be the roots of $3x^2 + 5x - 1 = 0$.

(i) State the values of $\alpha + \beta$, and $\alpha\beta$. 1

(ii) Form the new quadratic equation with roots $\frac{1}{\alpha^2}$ and $\frac{1}{\beta^2}$ 2

- c) The diagram shows a lighthouse at the top of a cliff. The combined height of the lighthouse and cliff is 200 m. A ship, A is due east of the lighthouse and the elevation of its top is 13° . A second ship, B finds that the lighthouse is on a bearing of 051°T , and the elevation of its top is 12° .



(i) Show that Angle $AQB = 141^\circ$ 1

(ii) Show that $AQ = 866.295$ m (correct to 3 decimal places) 1

(iii) Calculate how far apart the ships A and B are, giving your answer correct to the nearest metre. 2

e) Simplify $\frac{1}{(n-1)!} + \frac{n^3+1}{(n+1)!}$ 2

End of Test

Solutions to 2013 Extension 1
Preliminary Final.

1. $P(2) = 7$ D
 $\therefore P(-2) = 7$ as $P(x)$ is even

2. $3^{n+1} + 3^n = 3^n(3+1)$
 $\therefore \frac{3^{1001} + 3^{1000}}{4} = \frac{3^{1000}(3+1)}{4}$ B
 $= 3^{1000}$

3. $P(1) = 0 \quad \therefore 2+1+2+a = 0$ B
 $a = -5$

4. $\sin(A+B) = \sin A \cos B + \cos A \sin B$ C
 $\sin(A-B) = \sin A \cos B - \cos A \sin B$
 $\therefore \sin(A+B) - \sin(A-B) = 2 \cos A \sin B$

5. Trial the answers
 $18 \times 8 = 144$ C

6. (TRY) acts as one letter
 $\therefore$ Arrange 6 things, 2 are alike
 $\therefore \frac{6!}{2!}$ D

Q7 a) $x = \frac{kx_2 + lx_1}{k+l}$ $y = \frac{ky_2 + ly_1}{k+l}$ $\frac{k}{l} = \frac{3}{-2}$

$x = \frac{3 \times 1 + (-2) \times 3}{3 + (-2)}$ $y = \frac{3 \times 2 + (-2) \times 4}{3 + (-2)}$

$x = 9$ $y = 14$

$\therefore P(9, 14)$

b) $m_1 = 1$ $m_2 = 2$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{1 - 2}{1 + 2} \right|$

$\tan \theta = \frac{1}{3}$

$\theta = \tan^{-1}\left(\frac{1}{3}\right) = 18^\circ 26'$

c) i) $\frac{13!}{3! 2! 2!}$

as there are 3 A's
2 L's
2 R's

ii) Put the L's together 1 way
Put the R's together 1 way
Arrange 11 things $\frac{11!}{3!}$

iii) Put the vowels together AAALIE
Arrange 9 things $\frac{5!}{3!}$ ways

$\frac{9! \times 5!}{2! 2! 3!}$

d) i) ${}^{15}C_4 \times {}^{10}C_3 = 163800$ ways

ii) Put them both in
 ${}^{14}C_3 \times {}^9C_2 = 13104$ ways

$\therefore$ Not both in is $163800 - 13104$
 $= 150696$ ways

8 a)

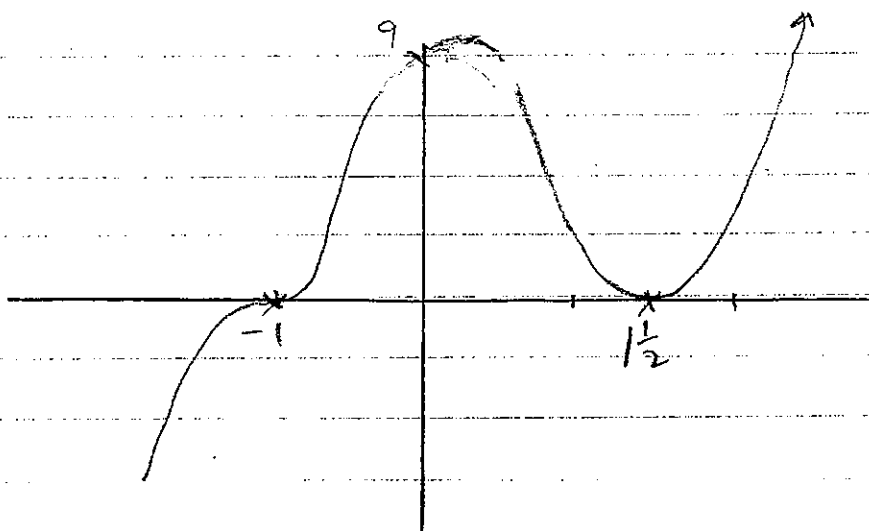
$$\begin{array}{r}
 3x^2 - 16x - 12 \\
 2x - 3 \overline{) 6x^3 - 41x^2 + 24x + 36} \\
 \underline{-6x^3 + 9x^2} \\
 -32x^2 + 24x + 36 \\
 \underline{+32x^2 - 48x} \\
 -24x + 36 \\
 \underline{-24x + 36} \\
 0
 \end{array}$$

$$P(x) = (2x - 3)(3x^2 - 16x - 12)$$

$$= (2x - 3)(3x + 2)(x - 6)$$

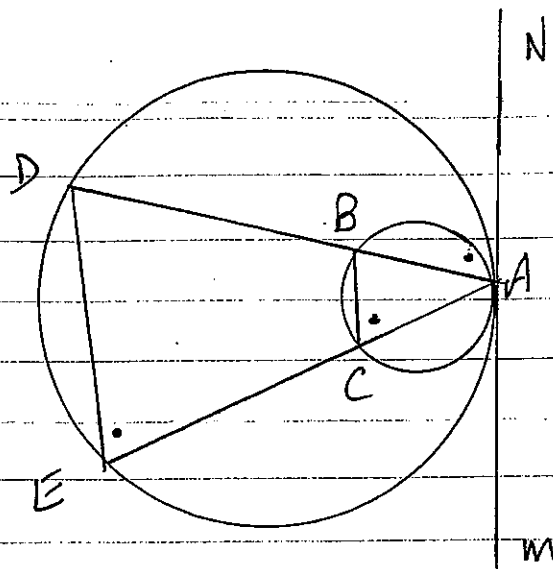
$$\begin{array}{r}
 3x + 2 \\
 x - 6
 \end{array}$$

b)



- c) $x = 80$ Angle at centre is twice angle at circumference standing on same arc.
- $y = 40$ Angles in the same segment standing on arc QR are equal.

d)



i)

In Δ 's ABC , ADE

1. $\angle A$ is common

2. $\angle NAB = \angle ACB$ (Angle between tangent + chord equals angle in alternate segment)

$\angle NAB = \angle AED$ (Angle between tangent + chord equals angle in alternate segment)

$\therefore \angle ACB = \angle AED$ (Both = $\angle NAB$)

$\therefore \Delta ABC \parallel \Delta ADE$ (2 angles test)

ii)

$\frac{AB}{AD} = \frac{AC}{AE} = \frac{BC}{DE}$ (corresponding sides in similar triangles)

or $\frac{AB}{AD} = \frac{BC}{DE}$ and (1)

$\frac{AC}{AE} = \frac{BC}{DE}$ (2)

(1) $\times$ (2)

$$\frac{AB}{AD} \times \frac{AC}{AE} = \frac{BC}{DE} \times \frac{BC}{DE}$$

$$\frac{AB \times AC}{AD \times AE} = \frac{BC^2}{DE^2}$$

9 a) i) $\cos(A+B) = \cos A \cos B - \sin A \sin B$

ii) put $A = 45^\circ$ $B = 30^\circ$

$$\cos 75^\circ = \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ$$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2}$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{6} - \sqrt{2}}{4}$$

b) LHS = $\frac{1 + \cos x + \sin x}{1 - \cos x + \sin x}$

$$= \frac{1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}}{1 - \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}}$$

$$\times \frac{1+t^2}{1+t^2}$$

$$= \frac{1+t^2 + 1-t^2 + 2t}{1+t^2 - 1 + t^2 + 2t}$$

$$= \frac{2+2t}{2t^2+2t}$$

$$= \frac{2(1+t)}{2t(t+1)} = \frac{1}{t}$$

$$= \cot \frac{x}{2}$$

$$c) \quad \sqrt{3} \sin \theta - \cos \theta = 1$$

check does $\theta = 180^\circ$

$$\begin{aligned} \text{LHS} &= \sqrt{3} \sin 180 - \cos 180 \\ &= 0 - (-1) \\ &= 1 = \text{RHS.} \end{aligned}$$

$\therefore \theta = 180^\circ$ is a solution.

$$\sqrt{3} \cdot \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2} = 1$$

$$2\sqrt{3}t - 1 + t^2 = 1 + t^2$$

$$2\sqrt{3}t = 2$$

$$\sqrt{3}t = 1$$

$$t = \frac{1}{\sqrt{3}}$$

$$\therefore \tan \frac{\theta}{2} = \frac{1}{\sqrt{3}}$$

$$\frac{\theta}{2} = 30^\circ$$

$$\theta = 60^\circ$$

$$\therefore \theta = 60^\circ \text{ or } 180^\circ$$

$$d) i) \quad \cos x + \sqrt{3} \sin x$$

$$\equiv A \cos(x - \alpha)$$

$$\equiv A \cos x \cos \alpha + A \sin x \sin \alpha$$

$$\therefore A \cos \alpha = 1 \quad + \quad A \sin \alpha = \sqrt{3}$$

$$\therefore \tan \alpha = \sqrt{3} \Rightarrow \alpha = 60^\circ$$

$$A^2 = 1^2 + (\sqrt{3})^2 \Rightarrow A = 2$$

ii)

$$2 \cos(x - 60^\circ) = 1$$

$$\cos(x - 60^\circ) = \frac{1}{2}$$

$$x - 60^\circ = 60^\circ \text{ or } 300^\circ \text{ or } -60^\circ$$

$$x = 120^\circ \text{ or } 360^\circ \text{ or } 0^\circ$$

10 a)

$$\frac{2x-1}{x} \geq x$$

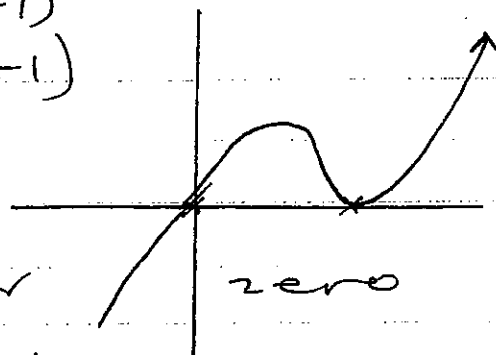
Note $x \neq 0$ Mult b/s by x^2 (which is positive)

$$x(2x-1) \geq x^3$$

$$0 \geq x^3 - x(2x-1)$$

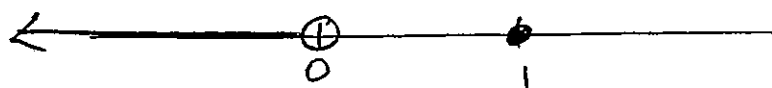
$$0 \geq x(x^2 - 2x + 1)$$

$$0 \geq x(x-1)^2$$



Need graph negative or

$$\therefore x < 0 \text{ or } x = 1$$



b)

$$3x^2 + 5x - 1 = 0$$

i)

$$\alpha + \beta = -\frac{5}{3}$$

$$\alpha\beta = -\frac{1}{3}$$

ii)

Sum of new roots

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$$

$$= \frac{\frac{25}{9} + \frac{2}{3}}{\frac{1}{9}} \times \frac{9}{9}$$

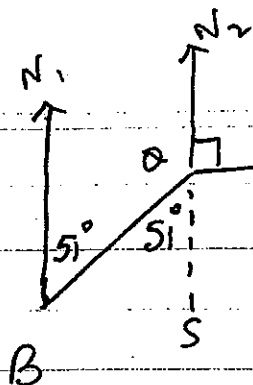
$$= 25 + 6 = 31$$

Product of new roots

$$\frac{1}{\alpha^2} \times \frac{1}{\beta^2} = \frac{1}{(\alpha\beta)^2} = 9$$

$$\therefore \text{Equation is } x^2 - 31x + 9 = 0$$

10. c) i)



$$\angle BQS = 51^\circ$$

(alternate $\angle$'s)

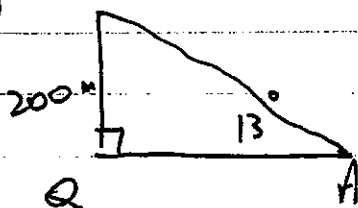
$$N_1 B \parallel N_2 S$$

$$\angle SQA = 90^\circ$$

(A is due East)

$$\therefore \angle BQA = 51 + 90 = 141^\circ$$

ii)



$$\tan 13^\circ = \frac{200}{QA}$$

$$\text{OR } \tan 77^\circ = \frac{QA}{200}$$

$$QA = \frac{200}{\tan 13^\circ}$$

$$QA = 200 \tan 77^\circ$$

$$QA \doteq 866.295 \text{ (to 3 decimal places)}$$

iii)

$$\tan 12^\circ = \frac{200}{BQ}$$

$$\text{OR } \tan 78^\circ = \frac{BQ}{200}$$

$$BQ = \frac{200}{\tan 12^\circ} \doteq 940.926 \text{ m}$$

$$BA^2 = QA^2 + BQ^2 - 2 \cdot QA \cdot BQ \cdot \cos 141^\circ$$

$$= 866.295^2 + 940.926^2 - 2 \cdot 866.295 \cdot 940.926 \cdot \cos 141^\circ$$

$$= 2902742$$

$$BA = 1703.74$$

$$BA \doteq 1704 \text{ m to nearest m}$$

e)

$$\frac{1}{(n-1)!} + \frac{(n+1)(n^2-n+1)}{(n+1)!}$$

$$= \frac{1}{(n-1)!} + \frac{n^2-n+1}{n!}$$

$$= \frac{n + n^2 - n + 1}{n!}$$

$$= \frac{n^2 + 1}{n!}$$